

CHI SQUARE TEST

The chi square test for Goodness of fit test claims about population proportions.

It is a non parametric test that is performed on categorical [ordinal and nominal] data.

There is a population of male who like different color bikes

	<u>Theory</u>	<u>sample</u>	(*) Goodness of fit test
Yellow Bike	$\frac{1}{3}$	22	
Red Bike	$\frac{1}{3}$	17	
Orange Bike	$\frac{1}{3}$	59	
	$\Downarrow$		
	Theory categorical distribution		Observed categorical distribution

Goodness of fit Test

In a science class of 75 students, 11 are left handed. Does this class fit the theory that 12% of people are left handed.

Ans →

	O	E
Left handed	11	9
Right handed	64	66
	<u>75</u>	<u>75</u>

$$\frac{312}{100} \times 75^3 = 9$$

O → observed

E → Expected